

WESTERN NEBRASKA, NE, USA

WMO: 725660

Lat: 41.871N

Lon: 103.593W

Elev: 1202

StdP: 87.69

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24028

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-20.7	-17.1	-24.1	0.5	-19.9	-20.8	0.7	-15.9	14.8	1.5	13.3	2.8	2.7	310	0.632

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.3	35.9	18.6	34.2	18.4	32.5	18.2	21.4	30.1	20.5	29.2	19.8	28.6	4.4	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	16.0	24.4	18.0	15.0	23.6	17.1	14.1	23.0	68.1	30.1	64.5	29.2	61.5	28.3	30.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.9	11.2	9.4	-26.2	38.9	4.3	1.2	-29.3	39.8	-31.8	40.4	-34.2	41.1	-37.4	41.9		
(5)			-26.4	24.1	4.3	2.4	-29.5	25.8	-32.0	27.2	-34.4	28.5	-37.5	30.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.7	-2.3	-0.9	4.1	8.2	13.7	20.1	23.8	22.4	17.3	9.4	2.9	-2.5		
	DBStd	10.58	6.24	6.40	5.80	5.05	4.74	4.09	2.91	2.93	4.67	5.06	5.50	6.45		
	HDD10.0	1670	382	305	193	91	19	1	0	0	5	71	217	388		
	HDD18.3	3574	639	537	441	305	154	28	2	6	73	278	463	646		
	CDD10.0	1577	0	1	10	36	136	303	429	386	223	51	3	0		
	CDD18.3	440	0	0	0	0	12	80	173	133	41	1	0	0		
	CDH23.3	5857	0	0	13	56	274	1113	2054	1561	692	92	2	0		
	CDH26.7	2770	0	0	1	7	89	511	1084	759	297	22	0	0		
(14) Wind	WSAvg	4.2	4.2	4.6	4.7	5.1	4.7	4.3	3.7	3.5	3.6	4.0	4.0	4.1		
(15) Precipitation	PrecAvg	392	11	13	26	42	65	66	51	29	28	24	16	13		
	PrecMax	630	32	49	67	99	184	168	122	88	107	77	55	39		
	PrecMin	196	0	0	4	7	7	15	2	1	1	2	0	0		
	PrecStd	104	8	12	18	25	42	41	33	22	24	18	14	9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.5	19.4	25.2	27.6	32.3	37.2	38.6	36.8	35.0	29.3	22.8	16.4		
		MCWB	6.9	7.7	10.4	13.0	16.0	18.8	18.7	17.8	17.7	14.1	9.8	7.2		
	2%	DB	12.6	15.5	21.6	24.9	29.1	34.4	36.6	34.8	32.5	26.0	19.0	13.0		
		MCWB	4.7	6.2	9.5	12.1	15.0	17.9	19.1	18.4	16.8	12.8	8.3	5.2		
	5%	DB	9.8	12.3	18.3	22.0	26.5	32.1	34.7	33.1	30.4	22.8	16.0	10.0		
		MCWB	3.5	4.6	7.8	10.8	14.3	17.8	18.9	18.6	16.2	11.6	7.0	3.7		
	10%	DB	7.0	9.1	15.1	18.9	23.6	29.6	32.7	31.2	27.6	19.5	12.7	7.1		
		MCWB	2.1	3.0	6.5	9.4	13.6	17.5	18.7	18.4	15.5	10.1	5.5	2.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.3	8.4	12.3	14.6	19.0	22.8	22.9	22.4	20.3	15.6	10.4	7.4		
		MCDB	14.7	18.2	21.7	24.7	27.2	32.0	31.8	30.8	30.4	25.7	20.9	15.3		
	2%	WB	5.3	6.5	9.8	12.6	16.7	20.4	21.6	21.0	18.7	13.6	8.8	5.5		
		MCDB	11.9	14.8	20.1	22.4	25.0	29.0	30.4	29.5	27.9	23.5	18.3	12.3		
	5%	WB	3.7	5.0	8.4	11.2	15.4	19.2	20.7	20.2	17.5	12.1	7.4	3.8		
		MCDB	9.2	11.7	17.2	20.8	23.5	28.2	29.6	28.6	26.5	20.4	15.6	9.5		
	10%	WB	2.2	3.3	6.9	9.9	14.3	18.0	19.9	19.3	16.5	10.8	5.8	2.2		
		MCDB	6.8	8.5	14.6	18.1	22.1	27.0	28.9	27.7	24.8	18.2	12.2	6.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.9	14.3	15.7	15.1	14.6	15.9	16.3	16.5	17.1	16.2	15.3	14.0		
		MCDBR	17.7	18.9	21.8	21.8	19.6	19.4	19.7	19.4	21.1	22.5	21.1	17.7		
		MCWBR	11.1	10.8	11.5	10.9	8.1	7.1	6.2	6.4	7.9	10.9	11.7	11.1		
	5% WB	MCDBR	16.6	18.2	20.1	20.1	17.0	17.4	16.5	16.6	18.3	19.7	20.0	16.8		
		MCWBR	10.7	10.7	11.2	10.5	7.8	7.9	6.2	6.5	7.6	9.9	11.3	10.9		
(40) Clear-Sky Solar Irradiance	taub		0.248	0.258	0.280	0.310	0.333	0.340	0.369	0.359	0.322	0.282	0.262	0.250		
	taud		2.580	2.546	2.513	2.432	2.439	2.445	2.401	2.415	2.498	2.579	2.555	2.551		
	Ebn at Noon		933	970	977	961	940	930	899	899	916	927	903	898		
	Edh at Noon		66	80	93	109	112	112	115	110	94	77	67	62		
(44) All-Sky Solar Radiation	RadAvg		2.25	3.11	4.34	5.33	6.17	7.29	7.19	6.28	5.07	3.49	2.46	1.88		
	RadStd		0.10	0.18	0.35	0.35	0.51	0.43	0.28	0.21	0.28	0.30	0.14	0.12		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (17 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page